

# Notes: Gopal and Schnabl (RFS, 2022)

## The Rise of Finance Companies and FinTech Lenders in Small Business Lending

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# 1 Big picture

- **Question.** Did the post-2008 decline in bank lending create a persistent shortage of credit for small businesses, or did finance companies and FinTech lenders substitute for banks?
- **Setting.** The paper studies U.S. secured, non-real-estate small business lending from 2006 to 2016. This is a market in which conventional data usually miss many nonbank lenders, so the paper builds a new dataset from Uniform Commercial Code (UCC) filings.
- **Main data contribution.** The authors collect public UCC filing records from all 50 states and Washington, DC. The final dataset contains about **11.26 million loans to 3.4 million unique firms**. It covers about **95% of secured non-real-estate small business lending** and about **73% of total small business lending**.
- **Main aggregate fact.** Bank lending dropped sharply after the 2008 financial crisis and did not return to its precrisis trend. Nonbank lending expanded instead. By 2016, nonbanks originated close to **60%** of new secured non-real-estate small business loans.
- **Who are the nonbanks?** The key nonbank groups are finance companies and FinTech lenders. Finance companies are the largest nonbank group, including captive finance arms of manufacturers and independent finance companies. FinTech lenders were tiny before 2010 but expanded rapidly afterward.
- **Core empirical idea.** Counties with a high precrisis bank market share were more exposed to the nationwide bank-lending shock. If nonbanks substitute for banks, nonbank lending should grow more in those high-bank-share counties. If total credit is fully replaced, total lending should not fall more in those counties.
- **Main lending result.** Moving from a county at the 10th percentile of precrisis bank share to one at the 90th percentile raises the postcrisis nonbank market share by about **6.9 percentage points** and raises nonbank lending by about **17.4%**. The estimated effect on total lending is essentially zero.
- **Demand-vs-supply problem.** A high precrisis bank share might also correlate with demand for specialized nonbank loans. The paper addresses this with industry fixed effects, collateral fixed effects, within-firm comparisons, three natural experiments, and a comparison between bank-owned and independent finance companies.
- **Real effects.** The replacement of bank credit by nonbank credit implies that the bank shock should have little long-run effect on local small-business activity. The paper finds no economically meaningful effect on establishment growth, employment growth, wage growth, or business expansion by 2016.
- **Takeaway.** The paper changes the interpretation of the postcrisis “missing bank lending” fact. The missing bank lending was largely replaced by finance companies and FinTech lenders. Small business credit supply became less bank-centered, and nonbanks played an important stabilizing role in the recovery from the 2008 crisis.

## 2 Data and measurement (key objects)

## 2.1 What is a UCC filing?

The data are based on filings under **Article 9 of the Uniform Commercial Code**. When a lender makes a secured business loan, it can file a UCC record to establish a security interest in collateral and to determine creditor priority in bankruptcy. The filing is similar in spirit to a mortgage record, but it applies to non-real-estate business collateral such as equipment, inventory, receivables, deposit accounts, chattel paper, and general intangibles.

The economic reason UCC filings are useful is simple:

$$\text{secured small business loan} \Rightarrow \text{strong incentive to file a UCC record.} \quad (1)$$

Without the filing, the lender is treated as unsecured and typically has lower recovery rights. Filing fees are small, and electronic filing is available. Thus the filings provide a broad view of secured small business lending, including lending by nonbanks that are missing from standard bank regulatory datasets.

## 2.2 Main UCC sample

The paper’s vendor-collected UCC data cover filings from all 50 states and Washington, DC between 2006 and 2016. The filings include:

- borrower name and address;
- lender name and address;
- filing date;
- collateral categories, with 34 broad types;
- DUNS identifier from Dun & Bradstreet;
- borrower industry, mainly SIC codes;
- borrower location, including county FIPS;
- firm characteristics available as of 2016.

The final sample restricts attention to nonfinancial, nonutility firms and excludes loans originated by government entities. The resulting dataset contains **11.26 million loans** to **3.4 million firms**.

## 2.3 Coverage and external validity

The key coverage logic is:

$$\text{UCC coverage} \approx \text{total small business lending} - \text{unsecured lending} - \text{real-estate-secured lending.} \quad (2)$$

The paper estimates that unsecured lending is about 5% of total small business lending and real-estate-secured lending is at most about 22%. Therefore, UCC data cover roughly:

$$1 - 0.05 - 0.22 = 0.73, \quad (3)$$

or about **73% of total small business lending**. The paper also estimates the total small business lending market in 2016 as about **\$957.7 billion**, using \$699.1 billion covered by UCC divided by the 73% coverage estimate.

This coverage is much broader than commonly used alternatives. Community Reinvestment Act data cover bank small business lending but miss many nonbanks and many smaller banks. SBA loan-level data cover only government-guaranteed loans. The UCC data therefore give a more complete view of the lender composition of small business credit.

## 2.4 Important data limitation: loan size is usually missing

A UCC filing generally records the existence of a secured loan but not the exact dollar size. The paper therefore uses the **extensive margin** of lending: loan counts and whether a firm receives a loan. This is not ideal, but the authors provide evidence that loan counts are highly correlated with loan volumes at the county-year level.

The paper also supplements the UCC data with Equipment Data Associates (EDA) information for equipment lending. EDA appends equipment values and allows the authors to estimate loan size for roughly one-third of the small business lending market.

## 2.5 Lender categories

The main lender categories are:

- **Banks:** commercial banks, credit unions, thrifts, and bank-owned nonbank subsidiaries in the baseline classification;
- **Finance companies:** nonbank lenders specializing in business and consumer credit;
- **Captive finance companies:** finance arms of manufacturers, such as equipment producers;
- **Independent finance companies:** finance companies not owned by banks or manufacturers;
- **FinTech lenders:** online or technology-based small business lenders, including lenders relying heavily on automated underwriting or online repayment data;
- **Other nonbanks:** cooperatives, investment companies, and nonfinancial firms providing trade credit.

A key institutional distinction is that finance companies compete with banks but do not issue deposits and therefore are not subject to bank regulation or deposit-insurance-based supervision. This difference matters because tighter bank regulation after the crisis may have constrained bank lending while leaving nonbank lenders freer to expand.

## 2.6 Outcome variables

The main outcome variables measure changes from 2007 to 2016:

$$\Delta \text{NonbankMarketShare}_c = \text{NonbankMarketShare}_{c,2016} - \text{NonbankMarketShare}_{c,2007}, \quad (4)$$

$$\Delta \text{NonbankLending}_c = \log(\text{NonbankLoans}_{c,2016}) - \log(\text{NonbankLoans}_{c,2007}), \quad (5)$$

$$\Delta \text{TotalLending}_c = \log(\text{TotalLoans}_{c,2016}) - \log(\text{TotalLoans}_{c,2007}). \quad (6)$$

The real-side outcomes are changes in establishments, employment, wages, and business expansion rates. These come from Census County Business Patterns, Census Statistics of U.S. Businesses, the Bureau of Labor Statistics, and Census population data.

## 2.7 Main treatment/exposure variable

The central exposure variable is the county's precrisis bank share:

$$\text{BankShare}_{06,c} = \frac{\text{Bank loans in county } c \text{ in 2006}}{\text{Total loans in county } c \text{ in 2006}}. \quad (7)$$

The interpretation is exposure to the nationwide bank-lending supply shock. If every bank pulls back by the same percentage, a county with a 90% initial bank share loses much more local credit supply than a county with a 10% initial bank share. This logic is similar to a shift-share or Bartik design, but the paper uses the initial bank share directly because it is more transparent.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Conceptual substitution equation

The paper's main economic mechanism can be summarized as:

$$\Delta \text{Total Credit}_c = \Delta \text{Bank Credit}_c + \Delta \text{Nonbank Credit}_c. \quad (8)$$

The postcrisis bank shock makes  $\Delta \text{Bank Credit}_c < 0$  in exposed counties. The key question is whether:

$$\Delta \text{Nonbank Credit}_c \approx -\Delta \text{Bank Credit}_c. \quad (9)$$

If yes, total credit is approximately unchanged, and long-run real effects should be small. If no, total credit falls and local small-business outcomes should deteriorate.

## 3.2 County-level baseline specification

The baseline regression is:

$$\Delta Y_c = \alpha_s + \gamma \text{BankShare}_{06,c} + \delta X_c + \varepsilon_c, \quad (1)$$

where  $\Delta Y_c$  is one of three outcomes: change in nonbank market share, growth in nonbank lending, or growth in total lending.  $\alpha_s$  denotes state fixed effects.  $X_c$  includes county controls for initial economic conditions and pretrends, such as unemployment, labor force participation, average wages, population, and pre-2006 growth in labor force participation, unemployment, establishments, and wages.

### Interpretation of $\gamma$ .

- If  $\gamma > 0$  when the outcome is nonbank market share, counties more exposed to banks experienced more nonbank substitution.
- If  $\gamma > 0$  when the outcome is nonbank lending growth, nonbanks expanded more in high-bank-share counties.
- If  $\gamma \approx 0$  when the outcome is total lending growth, nonbank growth offset the bank pullback.

The paper’s preferred Table 4 estimates are:

| Outcome                                | Nonbank share | Nonbank lending | Total lending |
|----------------------------------------|---------------|-----------------|---------------|
| Coefficient on BankShare <sub>06</sub> | 0.212         | 0.534           | 0.016         |
| Significance                           | ***           | ***             | n.s.          |
| Economic move, p10 to p90              | +6.9 pp       | +17.4%          | approx. 0     |

**Economic message.** Exposure to the bank shock predicts a large increase in nonbank lending but not a fall in total lending. This is the paper’s central substitution result.

## 3.3 County-industry and county-collateral specifications

To address the concern that high-bank-share counties simply demanded more specialized nonbank loans, the paper adds industry and collateral comparisons.

For county-industry cells:

$$\Delta Y_{j,c} = \alpha_s + \eta_j + \gamma \text{BankShare}_{06,c} + \delta X_c + \varepsilon_{j,c}, \quad (2)$$

where  $j$  indexes two-digit SIC industries and  $\eta_j$  are industry fixed effects.

For county-collateral cells, the same equation is estimated with collateral fixed effects instead of industry fixed effects. Loans with multiple collateral types are assigned to a main collateral category.

**Interpretation.** These specifications compare counties while holding constant the type of borrower industry or the type of pledged collateral. If the baseline result were driven only by specialized demand for equipment finance, retail receivables finance, or industry-specific credit, the coefficient on BankShare<sub>06,c</sub> should shrink substantially or disappear.

The results do not disappear. At the county-industry level, the 10th-to-90th percentile move in bank share implies a **3.6 percentage point** increase in nonbank market share and a **7.8%** increase in nonbank lending. At the county-collateral level, it implies a **6.5 percentage point** increase in nonbank market share and a **14.6%** increase in nonbank lending. In both cases, total lending effects are near zero.

### 3.4 Within-firm substitution specification

The strongest demand-control exercise focuses on firms that borrowed from both banks and nonbanks before the crisis. For these firms, the paper asks whether the same firm is more likely to receive a repeat loan from a nonbank rather than a bank after 2008:

$$\text{RepeatLoan}_{i,b} = \alpha_i + \beta \text{Nonbank}_b + \delta X_i + \varepsilon_{i,b}. \quad (3)$$

Here  $i$  indexes firms and  $b$  indexes lender type.  $\alpha_i$  are firm fixed effects. The coefficient  $\beta$  compares the same firm's postcrisis repeat borrowing from nonbanks versus banks.

**Interpretation.** Firm fixed effects absorb borrower demand, quality, business model, location, and other time-invariant firm characteristics. A positive  $\beta$  indicates that the same firm substituted from bank borrowing to nonbank borrowing.

The main coefficient is about **3.4 percentage points**. Firms with precrisis relationships with both lender types are more likely to receive a repeat nonbank loan than a repeat bank loan after the crisis.

### 3.5 Single-lender-type repeat-loan specification

The paper also studies firms that borrowed from either a bank or a nonbank in 2006–2007, but not both:

$$\text{RepeatLoan}_i = \alpha + \beta \text{Nonbank}_i + \delta X_i + \varepsilon_i. \quad (4)$$

The dependent variable equals one if the firm receives a repeat loan after 2008.  $\text{Nonbank}_i$  equals one if the firm's preperiod lender type was nonbank.

The estimated effect is positive. Depending on controls and weights, nonbank borrowers are about **2.9 to 9.4 percentage points** more likely to receive a repeat loan than bank borrowers. This supports the interpretation that nonbank credit availability was stronger after the crisis.

### 3.6 Natural-experiment specifications

The paper replaces the simple precrisis bank share with three bank-shock exposure variables:

$$\Delta Y_c = \alpha_s + \theta \text{ShockExposure}_c + \delta X_c + \varepsilon_c. \quad (5)$$

The three exposures are:

- **FAS 166/167 exposure:** county share of lending by banks forced to consolidate off-balance-sheet vehicles;
- **Stress-test exposure:** county share of lending by banks later subject to stress tests;
- **Capital-raise exposure:** county-level exposure to required Tier 1 capital increases after Dodd-Frank.

All three shocks increase nonbank lending but do not increase or decrease total lending in a meaningful way. This provides independent support for the supply-shock interpretation.



### 3.7 Real-effects specification

To study local real outcomes, the paper estimates:

$$\Delta Y_c = \alpha_s + \gamma \text{BankShare}_{06,c} + \delta X_c + \varepsilon_c, \quad (6)$$

where  $\Delta Y_c$  is growth in establishments, employment, wages, or the expansion rate. The paper also estimates a county-industry version.

**Interpretation.** If nonbanks fully offset missing bank credit,  $\gamma$  should be close to zero for real outcomes. This is what the paper finds. The absence of long-run real effects is consistent with the lending results.

### 3.8 Dynamic firm-level specification

The robustness section also estimates yearly firm-level dynamics:

$$\text{Loan}_{i,t} = \alpha_i + \delta_t + \gamma_t \text{BankShare}_{06,c} + \varepsilon_{i,t}. \quad (7)$$

Here  $\gamma_t$  measures whether firms in high-bank-share counties are more or less likely to receive new loans in year  $t$ . The pattern shows a short-run credit decline after the crisis that fades by 2014, consistent with nonbanks replacing bank lending over time.

## 4 Identification and credibility

### 4.1 What identifies the substitution effect?

The design uses precrisis bank market share as exposure to a nationwide negative bank-lending shock. The core identifying comparison is:

$$\text{high-bank-share county after crisis} \quad \text{vs.} \quad \text{low-bank-share county after crisis}. \quad (10)$$

The paper argues that the banking-sector shock was national: banks faced losses, tighter regulation, stress tests, changed risk models, and higher capital requirements. But the shock mattered more in counties where banks had been the dominant small-business lenders. This creates cross-county variation in the intensity of local exposure to the bank pullback.

### 4.2 Main identifying assumption

The key assumption is:

$$\text{BankShare}_{06,c} \not\rightarrow \text{postcrisis demand for nonbank credit except through bank-credit exposure}. \quad (11)$$

In words, counties with high bank shares in 2006 should not have had separate, unobserved postcrisis demand shocks that made firms want more nonbank credit for reasons unrelated to the bank shock.

### 4.3 Why the assumption is plausible

The paper gives several pieces of support:

- **County comparability.** Counties above and below the median bank share have similar precrisis unemployment, labor-force participation, wages, population, and pretrends.
- **State fixed effects.** The preferred specification compares counties within states, reducing concerns about state-level regulation or macroeconomic differences.
- **Industry controls.** Results remain when comparing firms in the same two-digit SIC industry.
- **Collateral controls.** Results remain when comparing loans backed by similar collateral.
- **Within-firm design.** Results remain when comparing the same firm's repeat borrowing from banks versus nonbanks.
- **Natural experiments.** Three independent bank-shock measures deliver the same result: nonbank lending rises and total lending is unchanged.
- **Bank-owned finance companies.** Bank-owned finance companies contract like banks, while independent finance companies expand. This points to bank regulation and balance-sheet constraints rather than borrower demand alone.

### 4.4 Why this is not just nonbank specialization

A natural alternative story is that finance companies and FinTech lenders expanded because firms wanted specialized credit products, not because banks pulled back. The paper's evidence is hard to reconcile with a pure demand story.

First, banks and nonbanks lend to broadly similar borrowers. They overlap in industries, firm size, and collateral, especially for independent finance companies. Second, nonbanks increasingly use collateral types previously associated with bank lending, such as deposit accounts and general intangibles. Third, the same firms are more likely to repeat-borrow from nonbanks after the crisis. Fourth, independent finance companies expand where banks pull back, whereas bank-owned finance companies do not.

### 4.5 Limitations

The paper is strong on loan counts and lender substitution, but several limitations matter for interpretation:

- **Loan size.** UCC filings generally do not report loan size, so most analysis uses loan counts. The paper validates this with EDA and CRA correlations, but loan counts are not the same as dollar credit supply.
- **Loan prices and terms.** The main data do not reveal interest rates, fees, covenants, or all contract terms. Nonbank loans may be more expensive or shorter maturity even if they replace bank loans in count.
- **Unsecured and real-estate lending.** UCC data miss unsecured lending and do not fully cover real-estate-secured lending.

- **Short-run versus long-run effects.** Total lending was weaker during the crisis period. The paper’s main conclusion is about substitution by 2016, not necessarily full immediate replacement in 2008–2010.
- **Welfare.** The paper shows that nonbanks filled the credit gap and that real outcomes did not fall by 2016. It does not fully evaluate whether nonbank credit was cheaper, safer, or more welfare-enhancing than bank credit.

## 4.6 Relation to the broader literature

The paper connects three literatures. First, it speaks to small-business credit and relationship lending by showing that nonbanks are major suppliers of small-business credit. Second, it relates to the rise of shadow banks and FinTech lenders in mortgage and syndicated loan markets. Third, it revises interpretations of postcrisis bank lending by showing that missing bank lending need not imply missing total credit.

## 5 Tables and figures

### 5.1 Figure 1: Number of new loan originations

**Read.** Panel A plots bank and nonbank loan originations from 2006 to 2016. Bank lending falls sharply after the crisis. Nonbank lending also dips during the crisis but then expands strongly. Panel B decomposes nonbanks into finance companies, FinTech lenders, and other nonbanks.

**Key numbers.** Bank lending drops by roughly 26–27% from 2007/2006 to 2010. Finance company lending grows strongly after 2010. FinTech lenders are nearly absent before 2010 but reach 78,543 loans by 2016.

**Economic message.** The market does not simply shrink after banks pull back. The composition of lenders changes, with nonbanks becoming dominant.

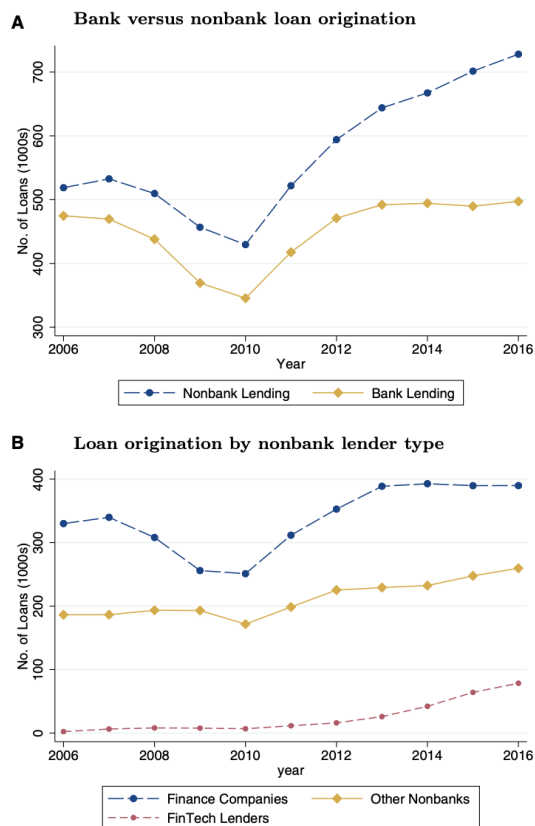
### 5.2 Table 1: Largest lenders

**Read.** The largest lenders are a mix of banks and finance companies. John Deere, CNH Capital America, Kubota Credit, GE Capital, and Caterpillar Financial Services appear alongside Wells Fargo, U.S. Bank, JPMorgan Chase, Bank of America, and BB&T.

**Economic message.** Finance companies are not fringe lenders in this market. Some are among the largest U.S. small-business lenders by UCC loan counts. This motivates treating them as central players rather than residual nonbank categories.

### 5.3 Table 2: Lender, borrower, and loan summary statistics

**Read.** From 2006 to 2016, banks make 4.96 million loans and nonbanks make 6.30 million. Nonbanks include 2.30 million captive finance loans, 1.41 million independent finance loans, and 269,905 FinTech loans. Nonbank borrowers are similar to bank borrowers in many dimensions, especially for independent finance companies.



**Figure 1**  
**Number of new loan originations**  
 This figure plots the number of loans originated annually between 2006 and 2016. Panel A shows total banks and nonbank lending. Bank lending captures loans originated by depository institutions (commercial banks, credit unions, thrifts, and nonbank subsidiaries of bank holding companies). Nonbank lending captures loans originated by all nondepository institutions. Panel B shows total lending for finance companies (sum of captive finance companies and independent finance companies), FinTech lenders, and other nonbanks (sum of cooperatives, investment companies and nonfinancials). Data on loan originations have been obtained from UCC filings.

**Table 1**  
**Summary statistics: Largest lenders**

| Lender                                       | Average annual loans | Bank | Total assets (2016, mil \$) | Loans outstanding (2016, mil \$) | Long-term debt (2016, mil \$) | Short-term debt (% of assets) |
|----------------------------------------------|----------------------|------|-----------------------------|----------------------------------|-------------------------------|-------------------------------|
| John Deere                                   | 54,586               | 0    | 34,064                      | 28,641                           | 16,568.80                     | 5388.9                        |
| Wells Fargo                                  | 28,112               | 1    | 1,227,235                   | 24,436                           | 232,061                       | 1,039,590                     |
| U.S. Bank                                    | 26,874               | 1    | 441,010                     | 10,818                           | 51,723                        | 343,142                       |
| CNN Capital America                          | 25,625               | 0    | 14,043                      | 12,823                           | 5,267                         | 560                           |
| Kubota Credit Corporation <sup>a</sup>       | 21,749               | 0    |                             |                                  |                               |                               |
| General Electric Capital (2014) <sup>b</sup> | 16,364               | 0    | 267,725                     | 33,759                           | 187,991                       | 25,000                        |
| JPMorgan Chase Bank                          | 15,620               | 1    | 2,082,803                   | 13,653                           | 397,216                       | 1,480,238                     |
| Caterpillar Financial Services               | 14,722               | 0    | 13,815                      | 26,212                           | 20,537                        | 7,094                         |
| Bank of America                              | 14,467               | 1    | 1,677,690                   | 29,370                           | 137,249                       | 1,334,032                     |
| Branch Banking and Trust                     | 14,052               | 1    | 214,433                     | 5,273                            | 18,340                        | 167,649                       |
| Scotiabank                                   | 9,874                | 0    |                             |                                  |                               |                               |
| Citizens Bank                                | 7,651                | 1    | 116,940                     | 2,461                            | 17,367                        | 83,233                        |
| De Lage Landen                               | 7,203                | 0    | 39,434                      | 28,525                           | 27,542                        | 4,796                         |
| Citibank                                     | 6,974                | 1    | 1,349,581                   | 7,930                            | 323,986                       | 880,790                       |
| AGCO Finance <sup>c</sup>                    | 6,952                | 0    | 2,278                       |                                  | 1610                          | 176                           |
| Bank of the West                             | 6,791                | 1    | 83,730                      | 2,468                            | 9,279                         | 62,277                        |
| TECF National Bank                           | 6,715                | 1    | 21,455                      | 2,373                            | 1,709                         | 17,390                        |
| Tower Loan <sup>a</sup>                      | 6,659                | 0    |                             |                                  |                               |                               |
| Toyota Motor Credit Corp <sup>c</sup>        | 6,408                | 0    |                             |                                  |                               |                               |
| PNC Bank                                     | 5,752                | 1    | 355,999                     | 4,030                            | 54,236                        | 262,381                       |

This table presents summary statistics for the 20 largest lenders in the sample. Average annual loans are the number of new loans per year from 2006 to 2016 based on UCC data. Bank is an indicator variable equal to one if the lender is a bank and zero otherwise. We collect the variables *Total assets*, *Total loans outstanding*, *Long-term debt*, and *Short-term debt* from 2016 financial statements, where available.  
<sup>a</sup> Private finance company with no financial reports filed.  
<sup>b</sup> GE Capital sold most of its financial arms and became a private corporation in October 2015. Statistics are as of 2014.  
<sup>c</sup> Filings consolidated to include both consumer and business loans. Breakdown is not available.

## Key numbers.

| Object                              | Banks        | Nonbanks     |
|-------------------------------------|--------------|--------------|
| Total loans, 2006–2016              | 4.96 million | 6.30 million |
| Average borrower employees, 2016    | 40           | 43           |
| Median borrower employees, 2016     | 6            | 6            |
| Average loan size, equipment sample | \$122,402    | \$132,902    |
| Average maturity                    | 36 months    | 28 months    |
| Equipment collateral share          | 65%          | 67%          |

**Economic message.** Banks and nonbanks serve overlapping small-business borrowers. The similarity is important because it makes substitution plausible.

**Table 2**  
Lender, borrower, and loan summary statistics

### A. Banks and nonbank lenders

This panel presents statistics on the number of lenders and the average number of loans originated by lender type.

|         | Number of lenders |              | Average number of loans |              | Total number of loans |
|---------|-------------------|--------------|-------------------------|--------------|-----------------------|
|         | ≥100 loans        | ≥1,500 loans | ≥100 loans              | ≥1,500 loans |                       |
| Bank    | 3,717             | 352          | 1,279                   | 10,112       | 4,958,852             |
| Nonbank | 3,841             | 408          | 1,416                   | 10,565       | 6,303,412             |

### B. Lenders by lender type

This panel presents the type of lenders in the sample. Column 1 presents the count of lenders with at least 1,500 loans in the sample. Columns 2 and 3 provide total lending and average number of loans by year for all lenders by category.

|                                     | Count (≥1,500 loans) | Total sample     | Avg per year   |
|-------------------------------------|----------------------|------------------|----------------|
| <b>Banks</b>                        | <b>352</b>           | <b>4,958,852</b> | <b>450,805</b> |
| Top-four banks                      | 4                    | 750,251          | 68,205         |
| Small banks: Covered by CRA         | 245                  | 2,756,304        | 250,573        |
| Other small banks and credit unions | 103                  | 1,383,681        | 125,789        |
| <b>Nonbanks</b>                     | <b>408</b>           | <b>6,303,412</b> | <b>573,037</b> |
| Captive finance                     | 92                   | 2,302,207        | 209,292        |
| Independent finance                 | 188                  | 1,407,878        | 127,989        |
| FinTech                             | 20                   | 269,905          | 24,537         |
| Cooperatives                        | 30                   | 301,944          | 27,449         |
| Investment companies                | 28                   | 118,848          | 10,804         |
| Other nonbanks                      | 50                   | 1,902,630        | 172,966        |

### C. Loan origination by lender type

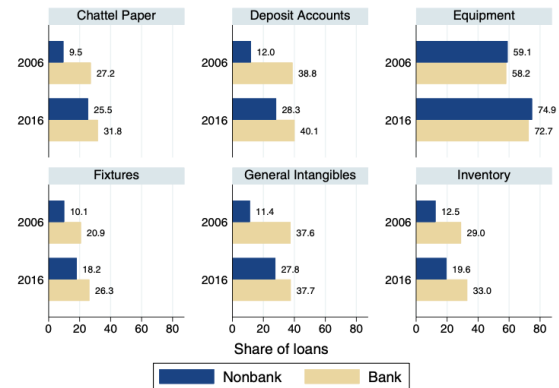
This panel presents the change in loan origination by lender type between 2006 and 2016.

|                                     | 2006           | 2010           | 2016           | Δ2006-10(%)   | Δ2010-16(%)  | Δ2006-16(%)  |
|-------------------------------------|----------------|----------------|----------------|---------------|--------------|--------------|
| <b>Banks</b>                        | <b>474,635</b> | <b>345,518</b> | <b>497,254</b> | <b>-27.20</b> | <b>43.92</b> | <b>4.77</b>  |
| Top-four banks                      | 85,120         | 52,081         | 73,651         | -38.81        | 41.42        | -13.47       |
| Small banks: Covered by CRA         | 263,673        | 196,389        | 284,852        | -25.52        | 45.04        | 8.03         |
| Other small banks and credit unions | 125,842        | 97,048         | 138,751        | -22.52        | 42.53        | 10.44        |
| <b>Nonbanks</b>                     | <b>518,669</b> | <b>429,489</b> | <b>727,937</b> | <b>-17.19</b> | <b>69.49</b> | <b>40.35</b> |
| Captive finance                     | 216,906        | 157,248        | 206,392        | -27.50        | 31.25        | -4.85        |
| Independent finance                 | 113,077        | 93,811         | 183,394        | -17.04        | 95.49        | 62.19        |
| FinTech                             | 2,326          | 6,807          | 78,543         | 192.65        | >> 100       | >> 100       |
| Cooperatives                        | 17,646         | 23,489         | 30,698         | 33.11         | 30.69        | 73.97        |
| Investment companies                | 11,378         | 6,864          | 11,569         | -39.67        | 68.55        | 1.68         |
| Other nonbanks                      | 157,336        | 141,270        | 217,341        | -10.21        | 53.85        | 38.14        |

### D. Borrower industry

This panel compares the likelihood of each borrower industry across the different lender types.

|                      | Full sample Mean | Banks Mean | Nonbanks Mean | Captive Mean | Independent Mean | FinTech Mean |
|----------------------|------------------|------------|---------------|--------------|------------------|--------------|
| Agriculture          | 0.151            | 0.105      | 0.187         | 0.281        | 0.090            | 0.026        |
| Construction         | 0.111            | 0.120      | 0.103         | 0.141        | 0.093            | 0.084        |
| Manufacturing/Mining | 0.115            | 0.113      | 0.117         | 0.101        | 0.119            | 0.074        |
| Retail trade         | 0.116            | 0.112      | 0.120         | 0.069        | 0.124            | 0.230        |
| Services             | 0.346            | 0.390      | 0.311         | 0.286        | 0.359            | 0.401        |
| Transportation       | 0.049            | 0.049      | 0.049         | 0.038        | 0.057            | 0.073        |
| Other                | 0.040            | 0.034      | 0.045         | 0.024        | 0.090            | 0.054        |
| Wholesale trade      | 0.072            | 0.076      | 0.068         | 0.060        | 0.067            | 0.059        |
| Obs.                 | 11,262,264       | 4,958,852  | 6,303,412     | 2,302,207    | 1,407,878        | 269,905      |



**Figure 2**

### Borrower collateral composition

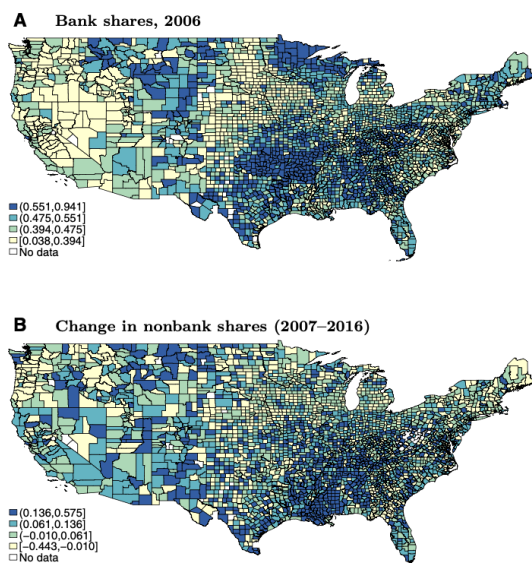
This figure depicts the share of bank and nonbank lending against the six most common types of collateral in our sample in 2006 and 2016. Shares can sum to greater than 100% because a single loan can have multiple types of collateral pledged. Data on loan originations and collateral pledged have been obtained from UCC filings.

## 5.4 Figure 2: Borrower collateral composition

**Read.** The figure compares collateral pledged to banks and nonbanks in 2006 and 2016. Both banks and nonbanks lend heavily against equipment. Over time, nonbanks increasingly take collateral categories that look more bank-like, including deposit accounts and general intangibles.

**Economic message.** Nonbanks are not merely expanding in one specialized niche. Their collateral mix converges toward banks', supporting the view that they can substitute for banks.

## 5.5 Figure 3 and Table 3: Bank share and county summary statistics



**Figure 3**  
**Geographical distribution of bank shares**  
Panel A maps the geographical distribution of bank shares in the United States in 2006. Bank share is the share of total county lending in 2006 originated by banks. Panel B maps the geographical distribution of change in nonbank market share between 2007 and 2016. Change in nonbank share is the change in the share of total county lending by nonbank lenders between 2007 and 2016. Data on loan originations have been obtained from UCC filings.

**Table 3**  
**County summary statistics**

|                                                 | Full Sample |         | Below Median |         | Above Median |         |
|-------------------------------------------------|-------------|---------|--------------|---------|--------------|---------|
|                                                 | Mean        | SD      | Mean         | SD      | Mean         | SD      |
| <i>Lending growth (2007–16)</i>                 |             |         |              |         |              |         |
| $\Delta(\text{Nonbank Lending})_{2007-16}$      | 0.298       | 0.507   | 0.214        | 0.470   | 0.382        | 0.528   |
| $\Delta(\text{Nonbank Market Share})_{2007-16}$ | 0.065       | 0.123   | 0.042        | 0.118   | 0.088        | 0.124   |
| $\Delta(\text{Total Lending})_{2007-16}$        | 0.171       | 0.434   | 0.143        | 0.406   | 0.199        | 0.459   |
| <i>Real outcomes (2007–16)</i>                  |             |         |              |         |              |         |
| $\Delta(\text{Employment})_{2007-16}$           | -0.020      | 0.149   | -0.020       | 0.150   | -0.021       | 0.148   |
| $\Delta(\text{Establishments})_{2007-16}$       | -0.056      | 0.094   | -0.057       | 0.092   | -0.055       | 0.096   |
| $\Delta(\text{Wages})_{2007-16}$                | 0.228       | 0.103   | 0.235        | 0.103   | 0.220        | 0.102   |
| $\Delta(\text{Expansion Rate})_{2007-14}$       | 0.004       | 0.046   | 0.004        | 0.048   | 0.004        | 0.044   |
| <i>Control variables</i>                        |             |         |              |         |              |         |
| Bank share <sub>2006</sub>                      | 0.476       | 0.130   | 0.376        | 0.077   | 0.576        | 0.086   |
| Top4 deposit share <sub>2005</sub>              | 0.094       | 0.162   | 0.109        | 0.180   | 0.078        | 0.140   |
| Population <sub>2005</sub>                      | 95,432      | 305,368 | 92,979       | 347,497 | 97,885       | 256,503 |
| Unemployment rate <sub>2005</sub>               | 5.436       | 1.761   | 5.381        | 1.858   | 5.491        | 1.658   |
| Labor force participation rate <sub>2005</sub>  | 0.491       | 0.062   | 0.495        | 0.062   | 0.486        | 0.061   |
| Average wage ('1,000s \$) <sub>2005</sub>       | 27.688      | 6.640   | 27.615       | 6.728   | 27.761       | 6.553   |
| $\Delta(\text{Labor force})_{2002-2006}$        | 0.031       | 0.070   | 0.028        | 0.070   | 0.034        | 0.070   |
| $\Delta(\text{Unemployment rate})_{2002-2006}$  | -0.812      | 1.154   | -0.829       | 1.172   | -0.794       | 1.136   |
| $\Delta(\text{Establishments})_{2002-2006}$     | 0.027       | 0.083   | 0.026        | 0.085   | 0.028        | 0.080   |
| $\Delta(\text{Wage})_{2002-2006}$               | 0.139       | 0.105   | 0.139        | 0.103   | 0.140        | 0.107   |
| Obs.                                            | 3,092       |         | 1,546        |         | 1,546        |         |

This table presents summary statistics at the county level. Median is based on 2006 county bank shares. County bank share represents the share of county lending that banks accounted for based on 2006 loan originations. Data on loan originations have been obtained from UCC filings.

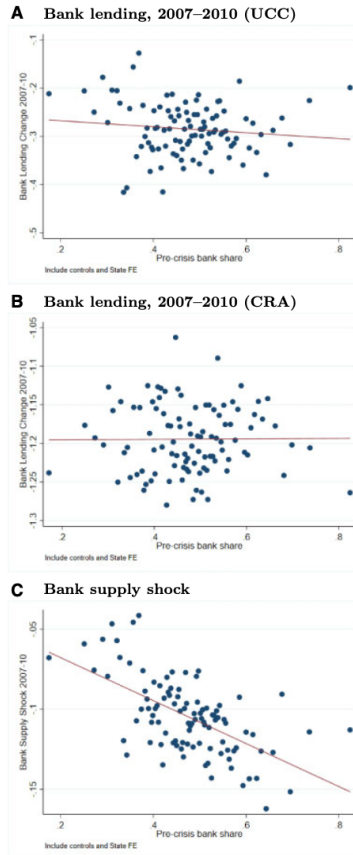
**Read.** Figure 3 maps bank shares in 2006 and changes in nonbank shares from 2007 to 2016. Table 3 shows that the average county bank share is 47.6%, with a standard deviation of 13.0%. Counties above and below the median bank share look similar in initial economic conditions and pretrends.

**Economic message.** The empirical design has meaningful cross-county variation in bank exposure, and high-bank-share counties do not look obviously different in basic precrisis economic conditions.

## 5.6 Figure 4: Bank lending change, 2007–2010

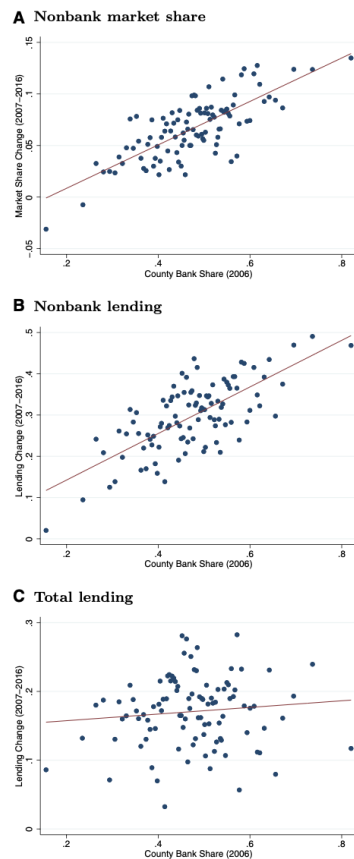
**Read.** Panels A and B show that the percentage decline in bank lending is not strongly related to the 2006 bank share. Panel C multiplies the bank lending decline by the initial bank share and therefore shows that the local credit-supply shock is much larger where banks were initially dominant.

**Economic message.** The shock is national, but local exposure differs. This is the logic behind using precrisis bank share as the exposure variable.



**Figure 4**  
**Bank lending change, 2007–2010**  
 This figure plots a binned scatter plot of the logarithm change in the number of bank loans in a county between 2007 and 2010 (panels A and B) and bank supply shock (panel C) on the regions' 2006 bank market shares as measured in the UCC data. Panel A plots the relationship in the UCC data, and panel B uses the CRA data. Bank supply shock (panel C) is the change in bank lending between 2007 and 2010 as a share of total lending in the region in 2007. Residuals are plotted after including state fixed effects and county controls. Controls include the county unemployment rate, labor force participation rate, average wage, and logarithm of the 2005 population, as well as the change in the unemployment rate, change in labor force participation rate, change in the number of establishments, and wage growth in the county between 2002 and 2006.

## 5.7 Figure 5 and Table 4: Baseline lending growth at the county level



**Figure 5**  
**Lending growth at the county level, 2007–2016**  
This figure presents binned scatter plots of change in nonbank market share (panel A), nonbank lending (panel B), and total lending (panel C) from 2007 to 2016 at the county level as a function of 2006 county bank shares. County bank share represents the share of county lending that banks accounted for based on 2006 loan originations. Change in nonbank market share is measured as the change in the share of total county lending by nonbanks. Lending changes are measured as the logarithm change in the number of loans. State fixed effects are included. Data on loan originations have been obtained from UCC filings.

**Table 4**  
**Lending growth at the county level**

|                                           | 2007–2016                   |                        |                      |
|-------------------------------------------|-----------------------------|------------------------|----------------------|
|                                           | (1)<br>Nonbank market share | (2)<br>Nonbank lending | (3)<br>Total lending |
| <i>A. Including State FE and controls</i> |                             |                        |                      |
| Bank share <sub>06</sub>                  | 0.212***<br>(0.024)         | 0.534***<br>(0.080)    | 0.016<br>(0.060)     |
| Controls                                  | Y                           | Y                      | Y                    |
| State FE                                  | Y                           | Y                      | Y                    |
| Obs.                                      | 3,021                       | 3,021                  | 3,021                |
| R <sup>2</sup>                            | .217                        | .521                   | .566                 |
| <i>B. Including controls</i>              |                             |                        |                      |
| Bank share <sub>06</sub>                  | 0.197***<br>(0.019)         | 0.645***<br>(0.073)    | 0.126**<br>(0.060)   |
| Controls                                  | Y                           | Y                      | Y                    |
| State FE                                  | N                           | N                      | N                    |
| Obs.                                      | 3,021                       | 3,021                  | 3,021                |
| R <sup>2</sup>                            | .0727                       | .12                    | .083                 |

This table presents county-level regressions for change in loan originations between 2007 and 2016. Change in nonbank market share is measured as the change in the share of total county lending by nonbanks. Lending change is measured as the logarithm change in the number of new loan originations and is winsorized at the 1% level. Bank share measures the share of bank lending in the county in 2006. Panel A includes state fixed effects. Controls include the county unemployment rate, labor force participation rate, average wage, and logarithm of the 2005 population, change in the unemployment rate in the county, change in labor force participation rate in the county, change in the number of establishments in the county, and wage growth in the county between 2002 and 2006. Standard errors are clustered at the county level.

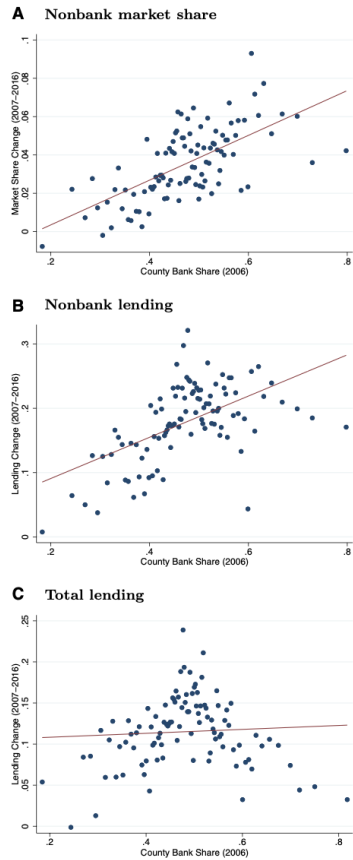
**Read.** Figure 5 shows binned scatter plots. High-bank-share counties experience larger nonbank market-share growth and larger nonbank lending growth, while total lending growth is flat. Table 4 estimates the same relationships using OLS with controls and state fixed effects.

### Key coefficients.

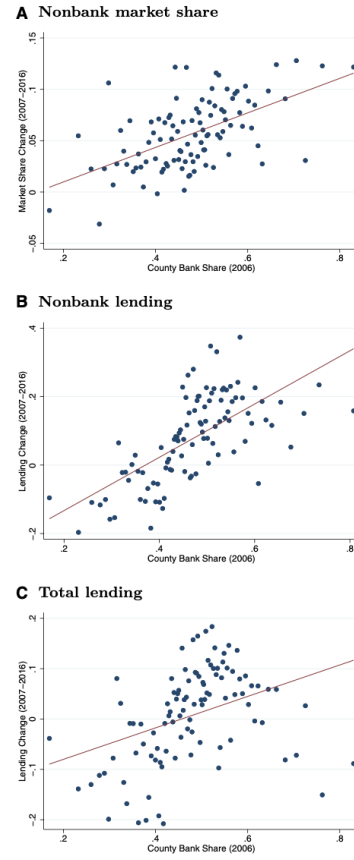
| Outcome                                    | Nonbank share | Nonbank lending | Total lending |
|--------------------------------------------|---------------|-----------------|---------------|
| BankShare <sub>06</sub> , with state FE    | 0.212***      | 0.534***        | 0.016         |
| BankShare <sub>06</sub> , without state FE | 0.197***      | 0.645***        | 0.126**       |

**Economic message.** This is the central evidence: nonbanks expanded precisely where banks had been important, and total lending did not fall more in those places.





**Figure 6**  
**Lending growth at the county-industry level, 2007–2016**  
 This figure presents binned scatter plots of change in nonbank market share (panel A), nonbank lending (panel B), and total lending (panel C) from 2007 to 2016 at the county-industry level as a function of 2006 county bank shares. County bank share represents the share of county lending accounted for by banks based on 2006 loan origination. Change in nonbank market share is measured as the change in the share of total county-industry lending by nonbanks. Lending changes are measured as the logarithm change in the number of loans. Two-digit SIC industry fixed effects (66 industries) are included. Data on loan originations have been obtained from UCC filings.



**Figure 7**  
**Lending growth at the county-collateral level, 2007–2016**  
 This figure presents binned scatter plots of change in nonbank market share (panel A), nonbank lending (panel B), and total lending (panel C) from 2007 to 2016 at the county-collateral level as a function of 2006 county bank shares. County bank share represents the share of county lending that banks accounted for based on 2006 loan originations. Change in nonbank market share is measured as the change in the share of total county-collateral lending by nonbanks. Lending changes are measured as the most common collateral in the sample. Collateral type fixed effects (23 types) are included. Data on loan originations have been obtained from UCC filings.

## 5.8 Figures 6–7 and Table 5: Industry and collateral controls

**Read.** Figure 6 and Panel A of Table 5 repeat the analysis at the county-industry level. Figure 7 and Panel B of Table 5 repeat it at the county-collateral level. In both cases, nonbank growth remains positively related to precrisis bank share and total lending remains essentially unchanged.

**Key coefficients.**

| Specification     | Nonbank share | Nonbank lending | Total lending |
|-------------------|---------------|-----------------|---------------|
| County-industry   | 0.111***      | 0.239***        | -0.066        |
| County-collateral | 0.201***      | 0.450**         | 0.005         |

**Economic message.** The substitution result is not only about particular industries or collateral types. It survives within finer borrower-product categories.

**Table 5**  
Lending growth at county-industry and county-collateral levels

|                                   | 2007-2016                   |                        |                      |
|-----------------------------------|-----------------------------|------------------------|----------------------|
|                                   | (1)<br>Nonbank market share | (2)<br>Nonbank lending | (3)<br>Total lending |
| <i>A. Two-digit SIC industry</i>  |                             |                        |                      |
| Bank share <sub>06</sub> (county) | 0.111***<br>(0.023)         | 0.239***<br>(0.068)    | -0.066<br>(0.052)    |
| Industry FE                       | Y                           | Y                      | Y                    |
| Controls                          | Y                           | Y                      | Y                    |
| State FE                          | Y                           | Y                      | Y                    |
| Obs.                              | 35,432                      | 35,432                 | 35,432               |
| R <sup>2</sup>                    | .0296                       | .16                    | .22                  |
| <i>B. Collateral</i>              |                             |                        |                      |
| Bank share <sub>06</sub> (county) | 0.201***<br>(0.050)         | 0.450**<br>(0.171)     | 0.005<br>(0.107)     |
| Collateral FE                     | Y                           | Y                      | Y                    |
| Controls                          | Y                           | Y                      | Y                    |
| State FE                          | Y                           | Y                      | Y                    |
| Obs.                              | 9,909                       | 9,909                  | 9,909                |
| R <sup>2</sup>                    | .166                        | .333                   | .321                 |

This table presents county-industry (panel A) and county-collateral (panel B) level regressions for change in loan originations between 2007 and 2016. Change in nonbank market share is measured as the change in the share of total county-industry lending by nonbanks. Lending change is measured as the logarithm change in the number of new loan originations and is winsorized at the 1% level. Bank Share measures the share of bank lending in the county in 2006. In panel A, industry is measured at the two-digit SIC level (64 industries) and includes two-digit industry and state fixed effects. In panel B, a loan is counted as belonging to a collateral category if the specified type of collateral is pledged in the loan. Loans with multiple collateral types are assigned to the most common (in sample) collateral type. Panel B includes collateral-type and state fixed effects. Controls include the county unemployment rate, labor force participation rate, average wage, and logarithm of the 2005 population, as well as the change in the unemployment rate in the county, change in labor force participation rate in the county, change in the number of establishments in the county, and wage growth in the county between 2002 and 2006. Standard errors are double clustered at the county and industry levels (panel A) and at the county and collateral levels (panel B).

**Table 6**  
Within-firm estimation

|                                                            | (1)                 | (2)                 | (3)                 | (4)                 |
|------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|
| <i>A. Firms with loans from both banks and nonbanks</i>    |                     |                     |                     |                     |
| Nonbank                                                    | 0.034***<br>(0.008) | 0.033***<br>(0.006) | 0.043***<br>(0.006) | 0.049***<br>(0.006) |
| Firm FE                                                    | Y                   | Y                   | N                   | N                   |
| County FE                                                  | N                   | N                   | Y                   | Y                   |
| Industry FE                                                | N                   | N                   | Y                   | N                   |
| Weighted                                                   | N                   | Y                   | Y                   | Y                   |
| Obs.                                                       | 298,350             | 298,350             | 298,350             | 298,350             |
| Cluster                                                    | County, Ind         | County, Ind         | County, Ind         | County, Ind         |
| R <sup>2</sup>                                             | .646                | .701                | .088                | .0312               |
| <i>B. Firms with loans from either a bank or a nonbank</i> |                     |                     |                     |                     |
| Nonbank                                                    | 0.029***<br>(0.008) | 0.052***<br>(0.008) | 0.090***<br>(0.010) | 0.094***<br>(0.013) |
| Firm FE                                                    | N                   | N                   | N                   | N                   |
| County FE                                                  | Y                   | Y                   | Y                   | N                   |
| Industry FE                                                | Y                   | Y                   | N                   | N                   |
| Weighted                                                   | N                   | Y                   | Y                   | Y                   |
| Obs.                                                       | 830,625             | 830,625             | 830,625             | 830,629             |
| Cluster                                                    | County, Ind         | County, Ind         | County, Ind         | County, Ind         |
| R <sup>2</sup>                                             | .0728               | .0979               | .0446               | .00918              |

This table presents changes in firm-level lending. The dependent variable is an indicator variable equal to one if the firm receives a repeat loan and zero otherwise. A loan is a repeat loan if the firm obtains a new loan from the same lender type in 2008–2016 as in 2006–2007 (preperiod). The sample in panel A is restricted to firms with both bank and nonbank loans in the preperiod. The sample in panel B is restricted to firms that had loans from only one type of lender in the preperiod. Nonbank is an indicator variable equal to one if the lender is a nonbank and zero otherwise. Industry fixed effects are at the four-digit SIC level. The regression is weighted by the number of loans to the firm from the lender type in 2006–2007 in columns 2–4. The mean of the dependent variable in panel A is 0.71 and in panel B is 0.52.

## 5.9 Table 6: Within-firm estimation

**Read.** Panel A examines firms that used both banks and nonbanks before the crisis. With firm fixed effects, these firms are 3.4 percentage points more likely to receive a repeat loan from a nonbank than from a bank. Panel B examines firms that used either a bank or a nonbank before the crisis; nonbank borrowers are more likely to receive a repeat loan afterward.

**Economic message.** The same firm's credit demand cannot explain the result. Firms shift toward nonbanks after the bank shock.

## 5.10 Table 7: Natural experiments

**Read.** The paper uses three separate bank supply shocks: FAS 166/167 exposure, stress-test exposure, and bank capital increases. Each shock raises nonbank market share and nonbank lending. None produces a meaningful effect on total lending.

## Key coefficients.

| Shock exposure    | Nonbank share | Nonbank lending | Total lending |
|-------------------|---------------|-----------------|---------------|
| FAS 166/167 share | 0.252***      | 0.561***        | 0.086         |
| Stress-test share | 0.209***      | 0.438***        | 0.086         |
| Capital increase  | 0.026***      | 0.065***        | 0.008         |

**Economic message.** The substitution result is not an artifact of the bank-share measure. Different bank supply shocks lead to the same pattern.

**Table 7**  
Natural experiments

|                                 | 2007-2016                   |                        |                      |
|---------------------------------|-----------------------------|------------------------|----------------------|
|                                 | (1)<br>Nonbank market share | (2)<br>Nonbank lending | (3)<br>Total lending |
| <i>A. FAS 166/167</i>           |                             |                        |                      |
| FAS share <sub>07</sub>         | 0.252***<br>(0.065)         | 0.561***<br>(0.178)    | 0.086<br>(0.144)     |
| Controls                        | Y                           | Y                      | Y                    |
| State FE                        | Y                           | Y                      | Y                    |
| Obs.                            | 3,025                       | 3,025                  | 3,025                |
| R <sup>2</sup>                  | .191                        | .509                   | .563                 |
| <i>B. Stress-tested banks</i>   |                             |                        |                      |
| Stress-test share <sub>07</sub> | 0.209***<br>(0.045)         | 0.438***<br>(0.143)    | 0.086<br>(0.105)     |
| Controls                        | Y                           | Y                      | Y                    |
| State FE                        | Y                           | Y                      | Y                    |
| Obs.                            | 3,025                       | 3,025                  | 3,025                |
| R <sup>2</sup>                  | .193                        | .509                   | .563                 |
| <i>C. Bank capital</i>          |                             |                        |                      |
| Δ(Cap) <sub>07-16</sub>         | 0.026***<br>(0.004)         | 0.065***<br>(0.011)    | 0.008<br>(0.008)     |
| Controls                        | Y                           | Y                      | Y                    |
| State FE                        | Y                           | Y                      | Y                    |
| Obs.                            | 3,025                       | 3,025                  | 3,025                |
| R <sup>2</sup>                  | .203                        | .514                   | .563                 |

This table presents county-level regressions of the absolute change in loan originations between 2007 and 2016 on FAS Share (panel A), stress test share (panel B), bank capital change (panel C). Lending change is measured as the change in the number of new loan originations and is winsorized at the 1% level. FAS Share measures the share of lending by banks affected by FAS166/167 in the county in 2007. Stress-test share measures the share of lending by stress tested banks in the county in 2007. Bank capital change is the weighted average of the percentage point change in bank Tier 1 risk-based capital in the county between 2007 and 2016 and is winsorized at 1%. The estimations include state fixed effects. Controls include the county unemployment rate, labor force participation rate, and average wage in 2005, as well as the change in the unemployment rate in the county, change in labor force participation rate in the county, change in the number of establishments in the county, and wage growth in the county between 2002 and 2006. Standard errors are clustered at the county level.

**Table 8**  
Bank-owned versus independent finance companies

|                          | 2007-2016                                         |                                                    |
|--------------------------|---------------------------------------------------|----------------------------------------------------|
|                          | (1)<br>Bank-owned finance company<br>market share | (2)<br>Independent finance company<br>market share |
| Bank share <sub>06</sub> | -0.033***<br>(0.010)                              | 0.043***<br>(0.017)                                |
| Controls                 | Y                                                 | Y                                                  |
| State FE                 | Y                                                 | Y                                                  |
| Obs.                     | 2,249                                             | 2,249                                              |
| R <sup>2</sup>           | .114                                              | .58                                                |

This table displays the results on lending changes between 2007 and 2016 at the county level. Change in market share is measured as the change in the share of total county lending by the lender type. Bank share measures the share of bank lending in the county in 2006. Bank-owned finance companies are wholly owned nonbank subsidiaries of bank holding companies. Independent finance companies are finance companies not owned by either manufacturers or banks. Controls include the county unemployment rate, labor force participation rate, average wage, and the logarithm of the 2005 population, as well as the change in the unemployment rate in the county, change in labor force participation rate in the county, change in the number of establishments in the county, and wage growth in the county between 2002 and 2006. Standard errors are clustered at the county level.

## 5.11 Table 8: Bank-owned versus independent finance companies

**Read.** In high-bank-share counties, the market share of bank-owned finance companies falls, while the market share of independent finance companies rises.

## Key coefficients.

| Outcome                 | Bank-owned finance company share | Independent finance company share |
|-------------------------|----------------------------------|-----------------------------------|
| BankShare <sub>06</sub> | -0.033***                        | 0.043**                           |

**Economic message.** If borrowers merely demanded finance-company products, bank-owned and independent finance companies should expand similarly. Instead, bank-owned finance companies behave more like banks. This supports the supply/regulatory channel.

## 5.12 Table 9: Real effects

**Read.** The bank-share exposure has no economically meaningful negative effect on establishment growth, employment growth, or wage growth. The expansion-rate coefficient is small and positive in

**Table 9**  
**Real effects, 2007–2016**

|                                 | (1)<br>$\Delta(\text{Establishments})$ | (2)<br>$\Delta(\text{Employment})$ | (3)<br>$\Delta(\text{Wages})$ | (4)<br>$\Delta(\text{Expansion Rate})$ |
|---------------------------------|----------------------------------------|------------------------------------|-------------------------------|----------------------------------------|
| <i>A. County level</i>          |                                        |                                    |                               |                                        |
| Bank share <sub>06</sub>        | −0.003<br>(0.016)                      | −0.036<br>(0.029)                  | −0.025<br>(0.018)             | 0.016*<br>(0.009)                      |
| Controls                        | Y                                      | Y                                  | Y                             | Y                                      |
| State FE                        | Y                                      | Y                                  | Y                             | Y                                      |
| Obs.                            | 3,033                                  | 3,033                              | 3,033                         | 3,033                                  |
| R <sup>2</sup>                  | .313                                   | .138                               | .211                          | .188                                   |
| <i>B. County-industry level</i> |                                        |                                    |                               |                                        |
| Bank share <sub>06</sub>        | −0.019<br>(0.026)                      | −0.058<br>(0.040)                  | −0.089<br>(0.058)             | 0.018**<br>(0.008)                     |
| Industry FE                     | Y                                      | Y                                  | Y                             | Y                                      |
| Controls                        | Y                                      | Y                                  | Y                             | Y                                      |
| State FE                        | Y                                      | Y                                  | Y                             | Y                                      |
| Obs.                            | 28,495                                 | 28,495                             | 28,495                        | 28,495                                 |
| R <sup>2</sup>                  | .23                                    | .112                               | .0999                         | .0754                                  |

This table presents results on the real effects of the substitution to nonbank lending. Bank share measures the share of bank lending in the county in 2006. Outcome variables are the logarithm of the change in the number of establishments (column 1), the logarithm of the change in the number of employees (column 2), the logarithm of the change in average wages (measured as the share of payroll per employee; column 3), and change in expansion rate (measured as the fraction of establishments that increase employment in a given year; column 4) between 2007 and 2016. The expansion rate is only available until 2014. Controls include the county unemployment rate, labor force participation rate, average wage, and the logarithm of the 2005 population, as well as the change in the unemployment rate in the county, change in labor force participation rate in the county, change in the number of establishments in the county, and wage growth in the county between 2002 and 2006. Standard errors are clustered at the county level in panel A. Panel B measures outcome variables at the county-industry (two-digit NAICS) level, and the standard errors are double clustered at the county and industry level.

some specifications.

**Economic message.** Because nonbanks replaced bank lending, the bank shock did not translate into persistent local small-business real effects by 2016.

## 5.13 Robustness exercises

**Read.** The robustness section shows that results hold across crisis and recovery periods, firm-level yearly dynamics, an IV implementation with a first-stage F-statistic of 26.6, controls for top-four bank share, population or lending weights, per capita outcomes, and controls for local retail mortgage bank share.

**Economic message.** The result is robust: the postcrisis lending gap is a bank-composition story rather than a total-credit-collapse story.

# 6 Short summary

## 6.1 One-paragraph version

Gopal and Schnabl build a new UCC-filing dataset covering secured non-real-estate small business lending from 2006 to 2016. They show that banks reduced lending after the 2008 crisis, but finance companies and FinTech lenders expanded and filled most of the gap. Their main design uses precrisis county bank share as exposure to the nationwide bank-lending shock. Counties with high bank shares see greater nonbank growth, but no corresponding fall in total lending. The result survives controls for industry and collateral, within-firm comparisons, three natural experiments, and a comparison between bank-owned and independent finance companies. Because nonbanks replaced bank credit, the paper finds no long-run negative effects on local establishments, employment, wages, or business expansion.

## 6.2 Five takeaways

1. Nonbanks were already important in small business lending before the crisis and became dominant afterward.
2. The decline in bank lending after 2008 did not imply a permanent decline in total small business lending.
3. Finance companies are the most important nonbank lenders; FinTech lenders grow rapidly from a tiny base.
4. The strongest evidence for substitution comes from high-bank-share counties, within-firm comparisons, and bank-shock natural experiments.
5. The paper shifts the policy question from “Did banks stop lending?” to “Who replaced banks, under what terms, and with what regulatory consequences?”

## 7 Conclusion

### 7.1 Core conclusion

The paper documents a major structural change in U.S. small business lending. After the 2008 financial crisis, banks reduced lending, but finance companies and FinTech lenders expanded. This nonbank expansion largely offset the missing bank lending by 2016.

### 7.2 Economic intuition

Banks faced postcrisis constraints: losses, stress tests, higher capital requirements, and tighter regulation. Nonbanks did not face the same bank-specific constraints. When banks pulled back in counties where they had been important lenders, nonbanks found lending opportunities and expanded. Because many nonbanks could lend against similar collateral and serve similar firms, they were able to substitute for banks rather than merely complement them.

### 7.3 Why the paper matters

The paper matters because much of the small-business-credit literature focuses on banks and relationship lending. This paper shows that such a bank-only view misses a large and growing part of the market. By 2016, nonbanks provided the majority of new secured non-real-estate small business loans. Any analysis of small business credit, financial regulation, or FinTech lending must therefore include finance companies and technology-based lenders.

### 7.4 Final takeaway

**The postcrisis lending gap was real for banks, but not for total small business credit.** The missing bank loans were largely replaced by finance companies and FinTech lenders, and this substitution helps explain why the bank-lending shock produced little long-run damage to local small-business outcomes by 2016.